

WEARABLE PIEZOELECTRIC ENERGY HARVESTER FOR HEALTH ANALYSIS FROM FOOT MOTION

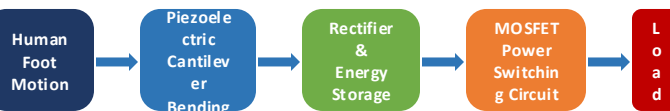
Problem Statement :

- Wearable biomedical devices rely on batteries requiring frequent charging
- Low-frequency human motion remains largely unused as energy source
- Need for compact, self-powered, and low-cost wearable energy solution

Objective :

- Design shoe-integrated piezoelectric energy harvester
- Harvest energy from human walking using LDT2-028K PVDF sensor
- Implement MOSFET switching for seamless power management
- Fabricate compact PCB integrating rectifier and switching circuit

Methodology



Experimental Results :

- Piezo open-circuit voltage: 0.7 V – 1.2 V (AC)
- Rectified DC output: ~0.5 V – 0.9 V (pulsating)
- Capacitor voltage (90s walking): ~1.8 V – 2.1 V
- Estimated harvested power: 170 – 220 μ W
- MOSFET switching threshold: 2.4 V (verified)
- PCB fabricated; switching circuit under debugging

Conclusion :

- Cantilever-based piezoelectric harvester produces 170–220 μ W from walking.
- MOSFET-based switching ensures uninterrupted load supply.
- PCB designed in KiCad; integration of step counter planned as future work.

System Overview



Fig 1: Piezoelectric Cantilever Prototype

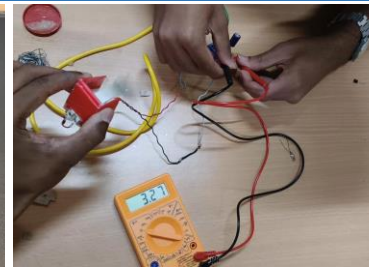


Fig 2: Breadboard Testing (3.27V)

Circuit Design & Simulation

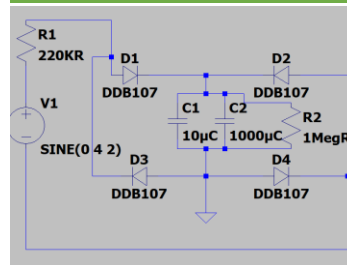


Fig 3: Rectifier & Switching Circuit (LTspice)



Fig 4: LTspice Simulation – Rectified DC Output (~1.8V)

MOSFET Switching Logic :

- Capacitor (2200 μ F) buffers harvested energy from walking
- $V > 2.4V \rightarrow$ Load powered by harvested energy (IRLML6402 ON)
- $V < 2.4V \rightarrow$ Auto-switch to rechargeable battery (800 mAh)
- PCB fabricated; switching section under active debugging

Working Principle

- Walking $\rightarrow$ low-frequency vibrations in shoe
- Vibrations deflect LDT2-028K cantilever beam
- PVDF piezo converts mechanical strain $\rightarrow$ AC voltage
- DB107 bridge rectifier converts AC $\rightarrow$ pulsating DC
- 2200 μ F capacitor stores & smooths rectified energy
- MOSFET switching circuit manages load power supply

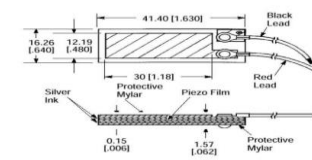


Fig 5: LDT2-028K Sensor



Fig 6: Oscilloscope Output (4.4V peak, 3.64 Hz)

Key Hardware Components

Component | Function

- | Component | Function |
|------------------------|----------------------------------|
| LDT2-028K (PVDF) | Piezo energy generation |
| DB107 Rectifier | AC to pulsating DC |
| 2200 μ F Capacitor | Energy storage & smoothing |
| IRLML6402 MOSFET | Power switching (2.4V threshold) |
| 800 mAh Battery | Backup power supply |

Future Work :

- Debug and verify MOSFET switching circuit on fabricated PCB
- Integrate step counter (MPU sensor) as wearable load
- Optimize cantilever dimensions for 1–2 Hz walking frequency
- Replace bridge rectifier with Schottky diodes to reduce drop losses